Name
Date
Course

Area (Part III): Regular Polygons, Regular Pyramids, and Geometric Probability (Homework)

Find the area of each regular polygon below with the given characteristics. Round to the nearest hundredth. Show all work.

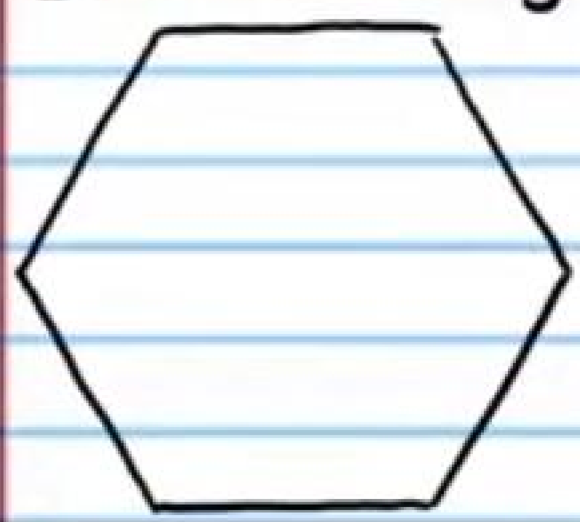
① Apothem: 5 ft

$\boxed{90.82\ ft^2}$

② Radius: 3 mm

$\boxed{25.46\ mm^2}$

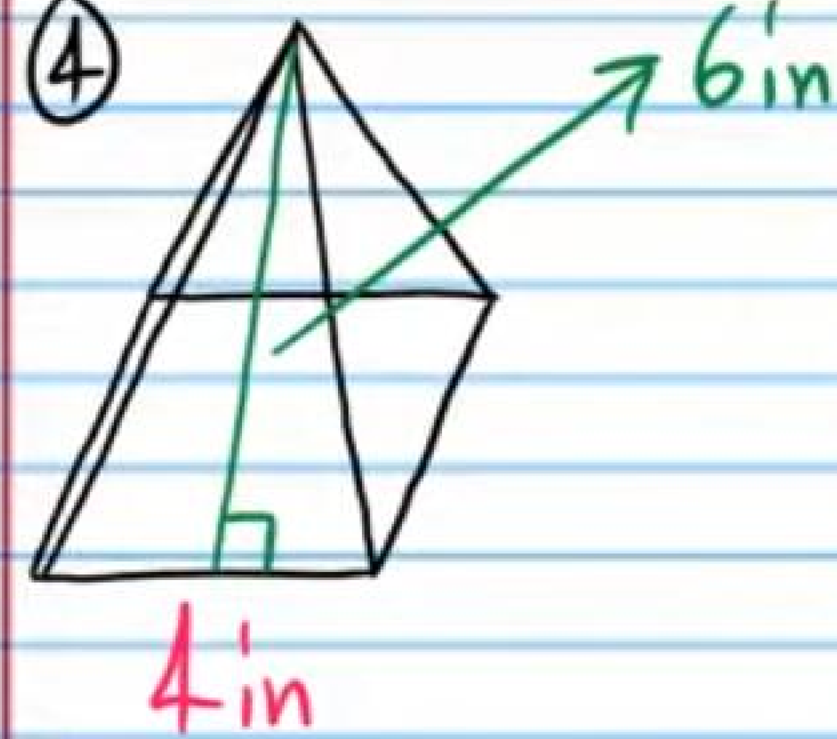
③ Side Length: 8 cm



$\boxed{166.28\ cm^2}$

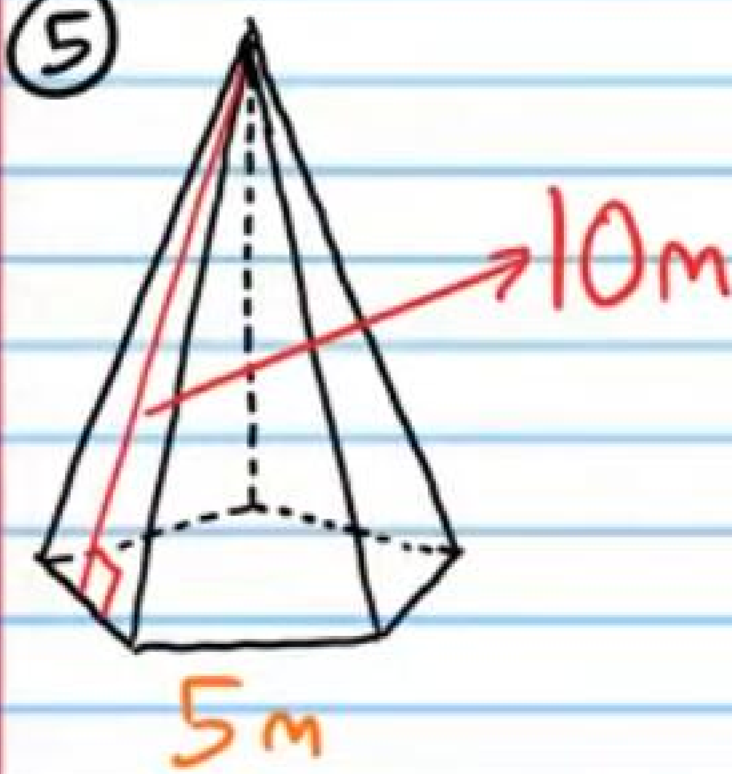
Find the surface area of each regular pyramid below. Round to the nearest hundredth if necessary. Carry extra numbers in intermediate steps to avoid round off error. Show all work.

④



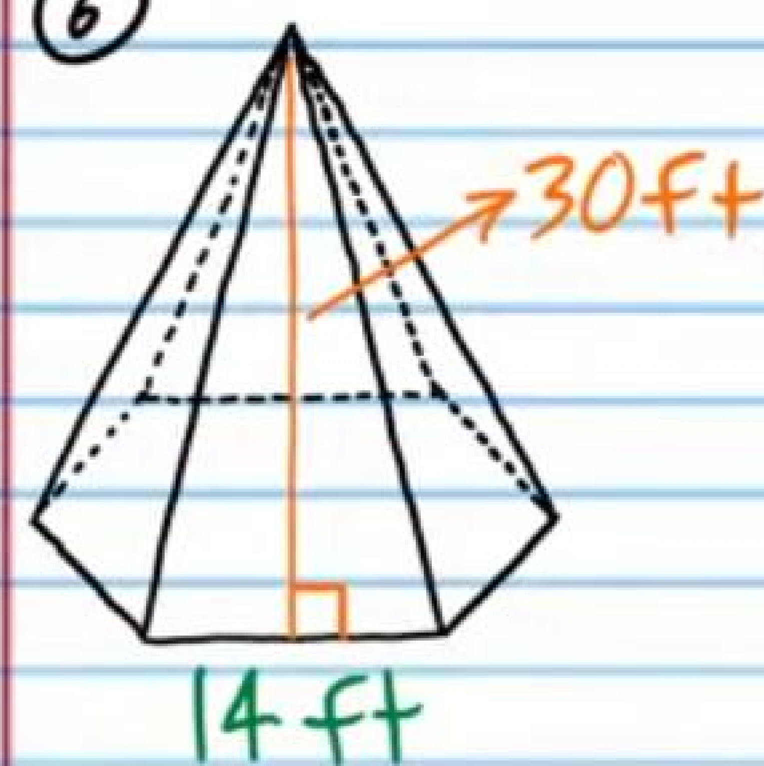
$\boxed{64\ in^2}$

⑤



$\boxed{168.01\ m^2}$

⑥

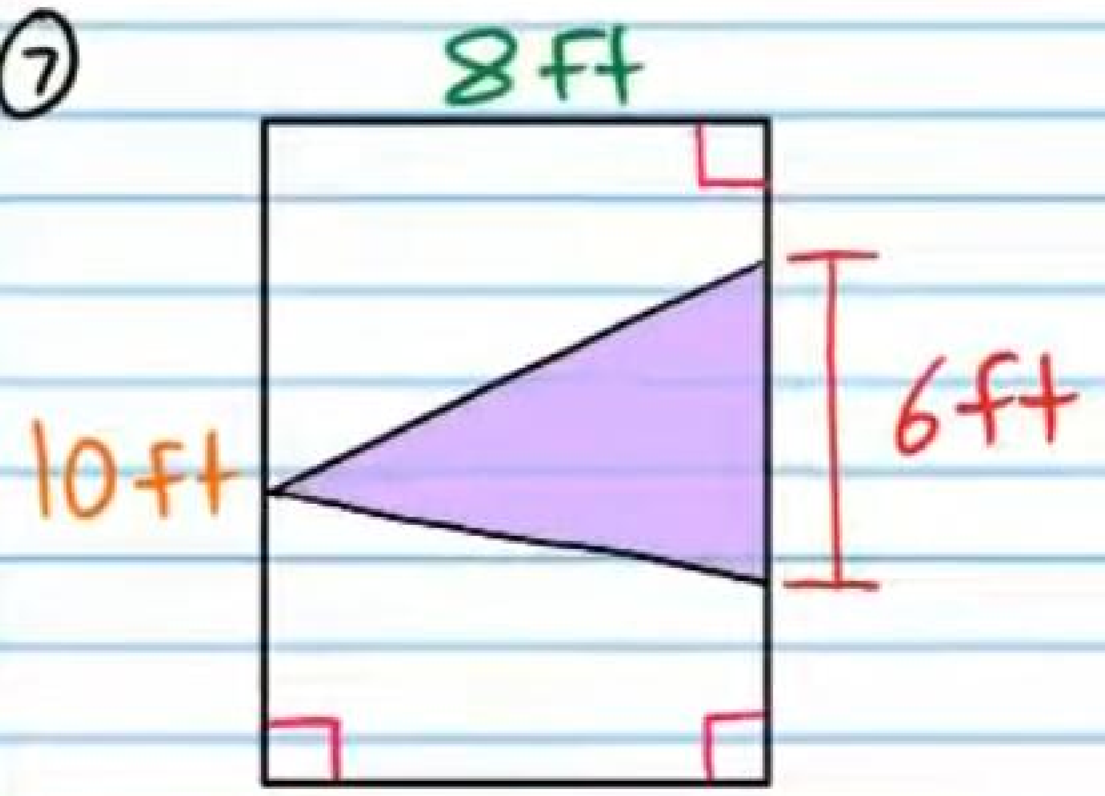


$\boxed{1,769.22\ ft^2}$

Find the probability that a randomly chosen point in the figure will lie in the shaded region. Round to the nearest thousandth if necessary.

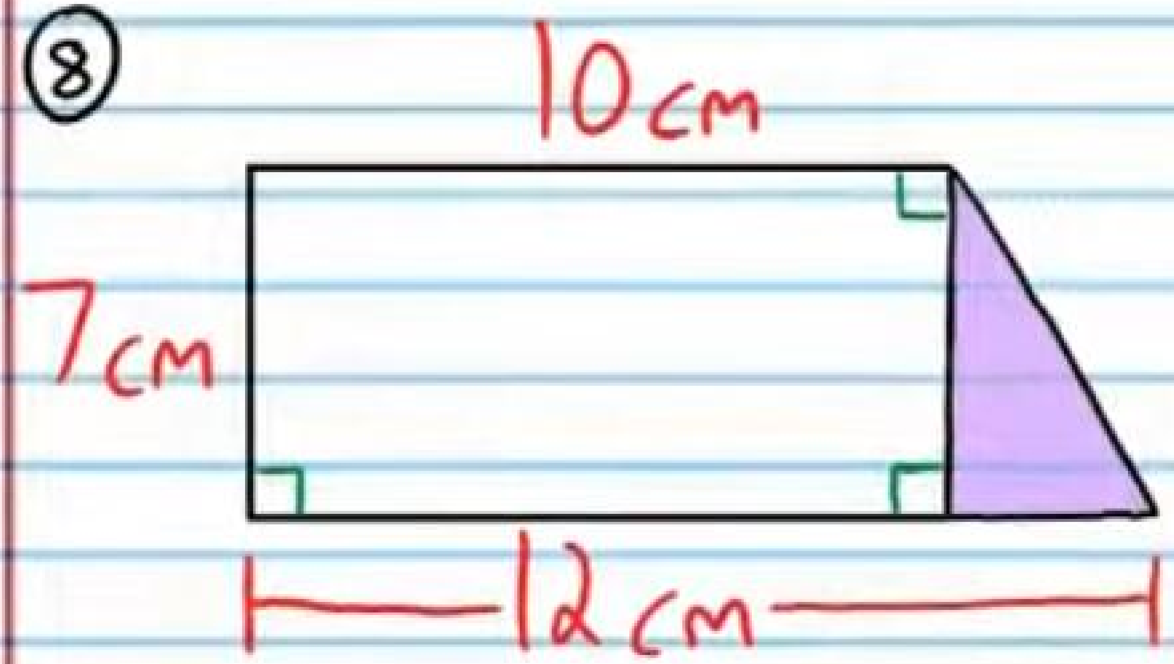
Show all work.

⑦



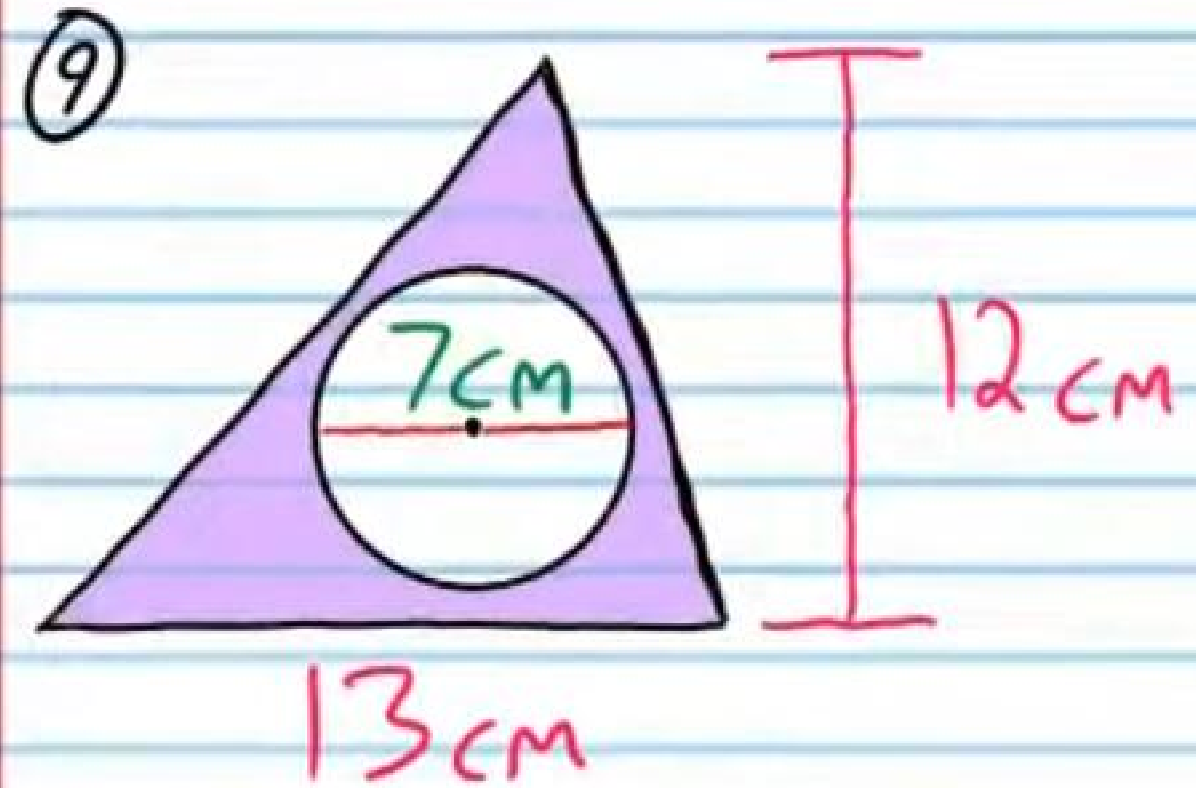
$\boxed{0.3}$

⑧



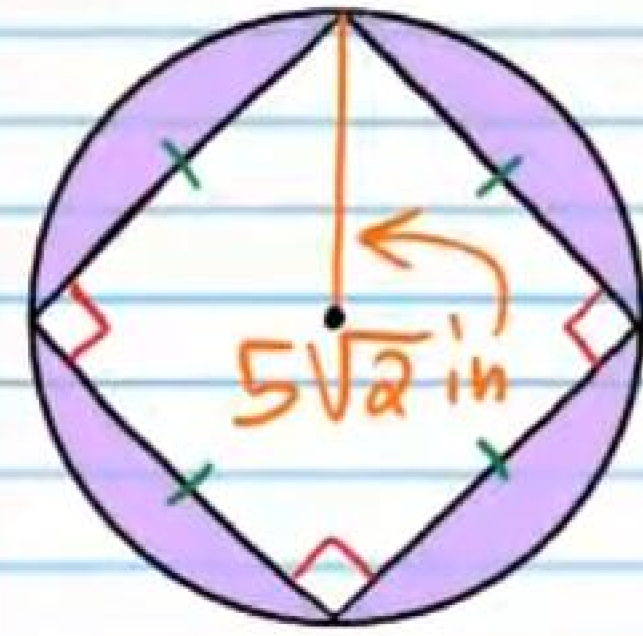
$\boxed{0.091}$

⑨



$\boxed{0.507}$

⑩ Hint: Use a 45°-45°-90° triangle.



$\boxed{0.363}$